RENEAL TANZANIA

2026 ICT Candidate Preparation Gap Analysis System

App Summary & Operational Report | September 2026

1. Executive Overview

The **Reneal Tanzania 2026 ICT Candidate Preparation Gap Analysis App** is a stats app designed to address a critical lab usage gap work: Given student candidate volume and actual 2025 logged lab hours, how large is the preparation gap per school, and where Reneal may take action.

The system evaluates **120 active government secondary schools** across 11–12 regions of Tanzania, managed by 8 technical field team. It tracks Form 2 (FTSEE) and Form 4 (CSEE) candidate results alongside non-exam (lagged) schools. The evaluation produces HTML web application featuring dynamic severity badges, filtering controls, visualizations, and browser-native PDF reporting.

2. Data Sources & Core Architecture

• **Lab Usage Data:** Derived from Reporter program script, recording active annual lab usage hours from 2019 to 2025. Schools with data logging failures in 2025 are displayed in italic grey with dash indicators to maintain systemic visibility without distorting averages.

• **Examination Results:** 2025 NECTA ICT exam results categorize candidates into three performance bands: A+B (Pass), C–D (Satisfactory/At-Risk), and F (Fail).

• **KNN Peer Benchmarking:** To establish realistic benchmarks ('Expected Hours'), a K-Nearest Neighbors formula identifies the 3 geographically closest exam-school peers using GPS Haversine distance. The target school's expectation is calculated as the mean actual usage of these 3 neighbors.

> CANDIDATE PREPARATION GAP FORMULA
>
> **Candidate Prep Gap (hrs/candidate) = (KNN Expected Hours − 2025 Actual Hours) ÷ ICT Exam Candidates**

The "expected hours" benchmark for each school is derived using a K-Nearest Neighbors (KNN) formula based on geographic proximity. For each school, the three geographically nearest exam schools are identified using the Haversine great-circle distance formula applied to GPS coordinates. The average of those three neighbors' actual usage hours is used as the expected benchmark for the target school.

KNN FORMULA

Expected Hours (school X) = Average actual hours of the 3 geographically nearest exam-school neighbors

Usage Gap = Expected Hours − Actual Hours

Gap Ratio = Gap ÷ Actual Hours (how many times over its own usage the school is behind its peers)

Note: Exam schools serve as KNN anchors. They do not receive their own Expected benchmark.

• **Preparation Floor:** Historical data establishes a baseline floor of **34 lab hours per candidate**. Falling below this threshold correlates with severe degradation in NECTA ICT examination performance.

• **Candidate Preparation Gap:** The Candidate Preparation Gap is the core metric for exam schools. It converts the raw hours shortfall into a per-candidate figure.

CANDIDATE PREP GAP FORMULA

Step 1: Usage Gap = KNN Expected Hours − 2025 Actual Hours

Step 2: Candidate Prep Gap = Usage Gap ÷ ICT Exam Candidates

Positive value → School is below its peer benchmark (under-prepared per candidate)

Negative value → School outperformed its KNN peers (above benchmark per candidate)

Null / — value → No 2025 usage data; gap cannot be calculated

Example: A school logged 800 actual hours against a KNN expected figure of 2,200 hours and had 20 candidates. The usage gap is 1,400 hours. The candidate prep gap is 1,400 ÷ 20 = 70 hrs/candidate — rated Critical.

• **Usage Gap for Non-Exam Schools:** Non-exam (lagged) schools did not enter candidates for NECTA ICT in 2025, so there is no candidate count against which to compute a per-candidate metric. Instead, Tab 3 uses a raw usage gap in hours and a gap ratio to characterize each school's performance relative to its KNN peer average.

NON-EXAM USAGE GAP FORMULA

Gap (hrs) = KNN Expected Avg (hrs/yr) − Actual Avg (hrs/yr)

Gap Ratio = Gap ÷ Actual Avg Hours

Positive gap → School is below its peer benchmark

Negative gap → School outperforms all 3 KNN neighbours

Gap Ratio > 1.0 → The school would need to more than double its usage to reach its benchmark

3. System Architecture & Tab Structure

Tab Name	Target Scope	Schools	Primary Operational Metric
Tab 1: Form 2 Exam	Schools sitting 2025 FTSEE ICT	38	Candidate Prep Gap (hrs/cand)
Tab 2: Form 4 Exam	Schools sitting 2025 CSEE ICT	23	Candidate Prep Gap (hrs/cand)
Tab 3: Non-Exam	Lagged schools without exam entries	76	Usage Gap (Expected − Actual)
Tab 4: Summary	Cross-tab executive analytics	120	System aggregates & team portfolios

4. Severity & Usage Classifications

Exam School Candidate Prep Gap Tiers

• **Critical (Red):** > 50 hrs/candidate shortfall. Severe under-preparation; immediate scheduling intervention required before exams.
• **High (Orange):** 20 – 50 hrs/candidate shortfall. Significant gap requiring a targeted lab scheduling allocation.
• **Moderate (Yellow):** 5 – 20 hrs/candidate shortfall. Below peer baseline but recoverable through structured access.
• **On Track (Green) / Above Benchmark (Blue):** 0 – 5 hrs shortfall (On Track) or negative gap (Above Benchmark outperforming peers).

Absolute Annual Usage Tiers

• **Good:** ≥ 2,000 hrs/yr logged. Active, regular laboratory utilization.
• **Moderate:** 500 – 1,999 hrs/yr logged. Lab in use but below optimal instructional frequency.
• **Poor / No Data:** < 500 hrs/yr logged (Poor) or zero logged hours due to system logging failure (No Data).

5. Key Cycle Findings (2025 Evaluation)

• **Form 2 vs. Form 4 Disparity:** Form 2 exam schools (1,404 candidates across 38 schools) averaged a gap of **12.74 hrs/candidate**. In contrast, Form 4 exam schools (371 candidates across 23 schools) recorded an average gap of **40.80 hrs/candidate** — over 3.2 times higher than Form 2.
• **Shared-Lab Bottleneck:** Of the 23 Form 4 exam schools, 18 share a single 20-client computer lab with the Form 2 cohort. Smaller Form 4 student numbers, combined with unmanaged lab sharing, lead to severe per-candidate time deficits.
• **Non-Exam Outliers:** 21 non-exam schools logged usage higher than their KNN exam-school benchmarks (e.g., Florian SS, Lucas Malia SS). These represent prime candidates for enrollment into the NECTA exam program.
• **Data Infrastructure Gaps:** 12 distinct schools (14 total cohort entries) recorded no usage data in 2025 due to different reasons, concentrated primarily in the Arusha/Longido corridor.

6. Workflows

1. Field Team: Filter SPA views by team portfolio. Prioritize immediate lab schedule adjustments for Critical/High gap schools. Enforce double-shift rotation schedules in shared-lab environments and execute technical site visits to restore logging at 'No Data' locations.
2. Program Workload: Utilize Tab 4 (Summary) to evaluate workload distribution across all 8 field team portfolios, ensuring resources are deployed to areas with the highest candidate prep risk.
3. Report Detail: Leverage one-page exported PDF summary reports for donor and engagement. Key headline metric: Reneal supported 1,775 candidates across 61 exam schools in 2025, achieving a 17.8% combined A+B pass rate despite structural lab sharing constraints.